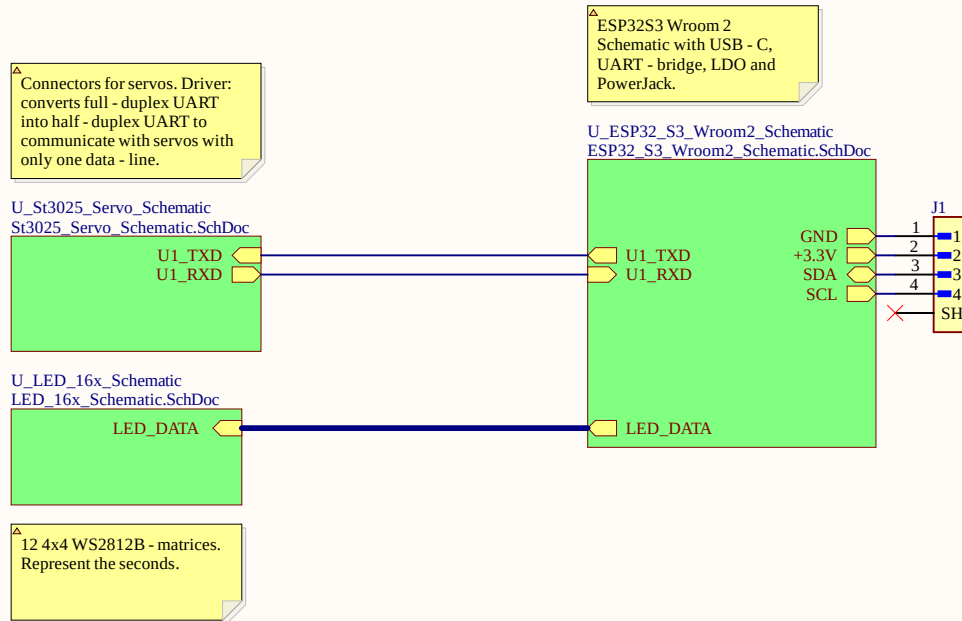
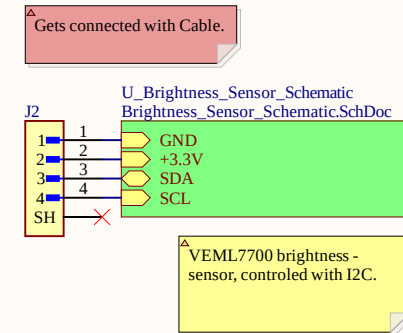


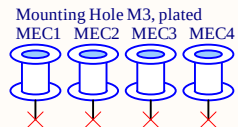
Power - Supply



Brightness - Sensor - Board



Mounting Holes



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Title of drawing: *Blackbox_BauhallenUhr_Schematic.SchDoc*

Date: 29.07.2026 Time: 16:16:58

Revision: V0.1

Drawn by: Manuel Hueppi

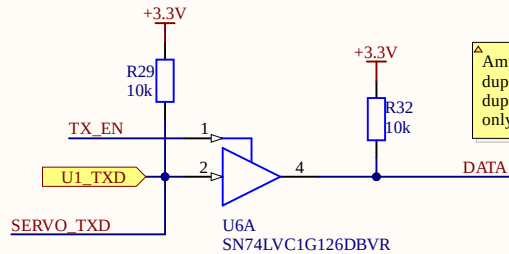
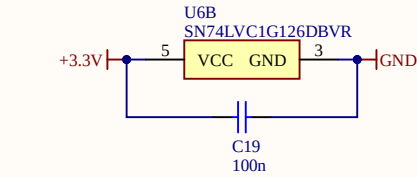
Institut: IBK
Departement: DBAUG

Project: *BauhallenUhr.PrjPcb*

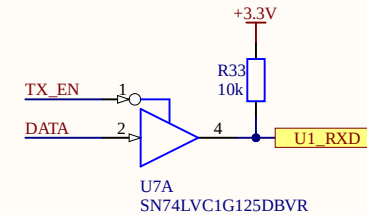
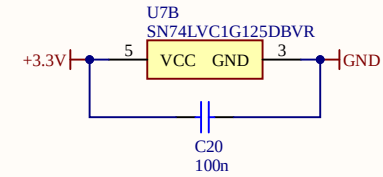
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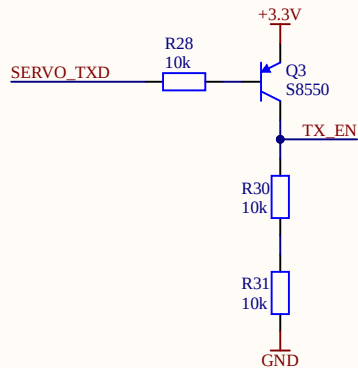
Single Bus Buffer Gate



Inverted Bus Buffer Gate



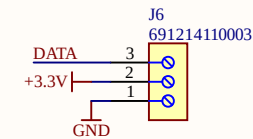
Half - Duplex Direction Detection



△ If the ESP32 writes something to the servos, the TX_EN - signal will enable the data flow between the SERVO_TXD - line and the DATA - line. If the ESP32 writes nothing, the data flow in the opposite direction is enabled, which means the ESP32 can read e. g the position of the servos.

Servo Connector

△ Only one connector, because these servos are controlled via UART, they can be chained together (serial).



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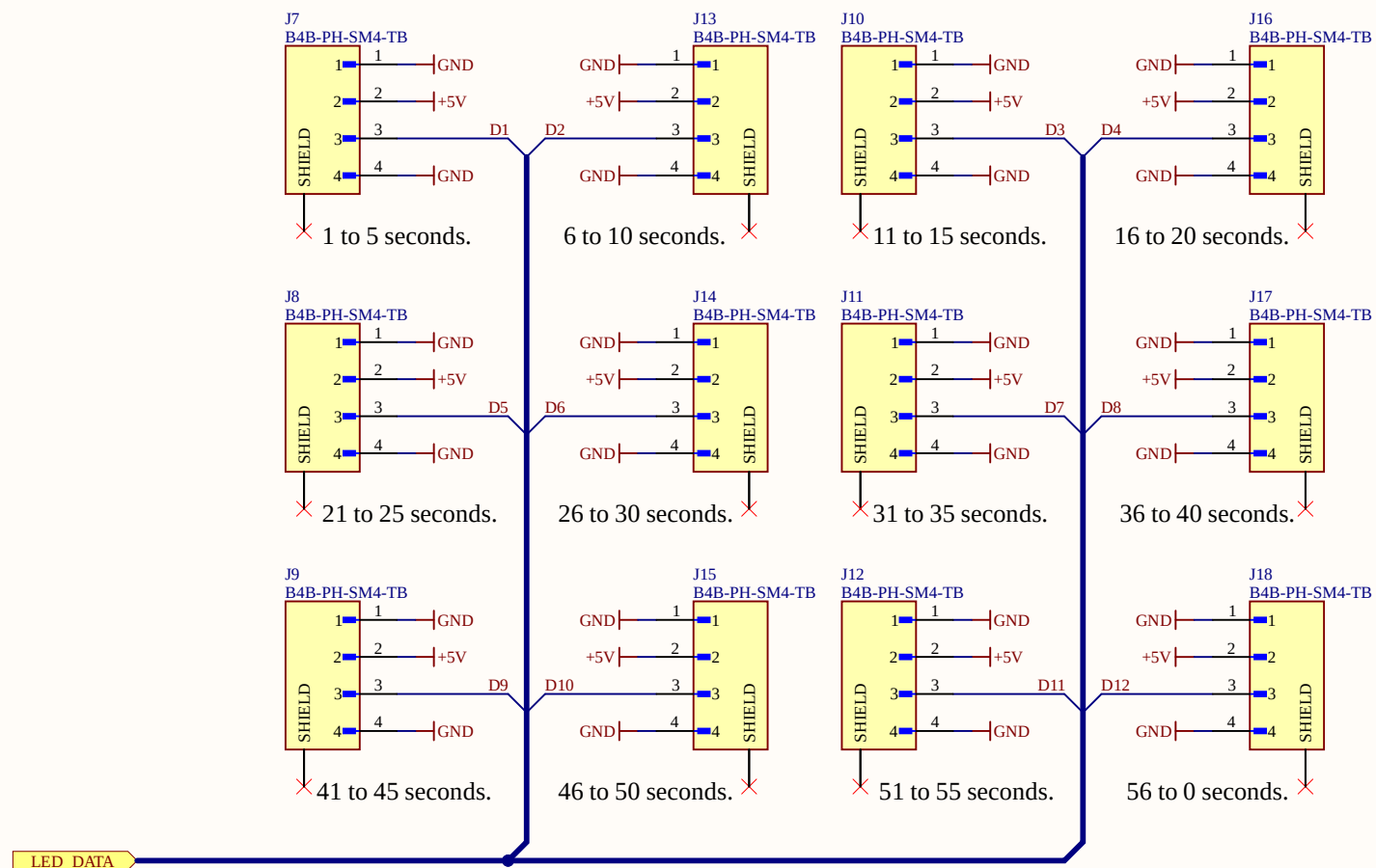
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Revision: V0.1

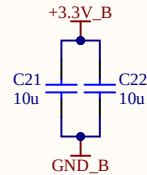
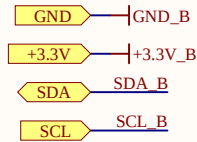
Drawn by: Manuel Hueppi
Institut: IBK
Departement: DBAUG

Project: BauhallenUhr.PrjPcb
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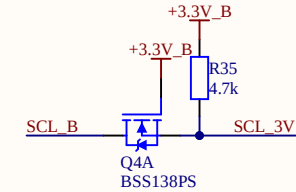
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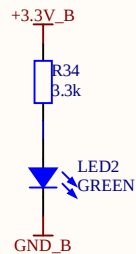
I2C Connector



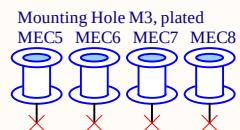
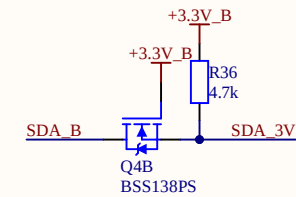
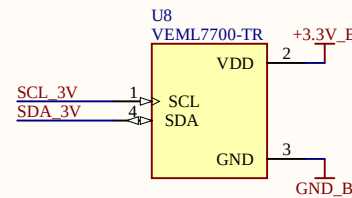
Shift with Pull - Ups



ON - LED



Brightness - Sensor



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Board Stack Report